

AFFORDABLE SENSOR-ENABLED MEDICAL TRAINING MANIKIN

ECOLOGY, ENVIRONMENT AND SUSTAINABILITY

Apr 2026

Contributors

Ankit Sinha (CS24B056), Ankit Prakash (CE24B002), Anirudh Krishnan (CE24B001), Akshith Asundi (PH24B001), Sahasvat R (CS24B038), Dhruv Pokhriyal (CS24B010), K T Rahul (EE24B018), Lakshay Dhyani (EE24B062), Purnima Jayakumar (CH24B028), Mritunjay Tiwari (CS24B054), Ganesh Rathod (ME24B031), Pradeep Kumar (ME24B053)

Abstract

Limited access to affordable medical training equipment and the absence of objective assessment methods constrains effective learning of critical emergency procedures. This project presents a low-cost, sensor-enabled medical training manikin integrated with a mobile application to enable data-driven, hands-on skill development. The system supports essential interventions, including cardiopulmonary resuscitation (CPR), rescue breathing, compression force and pulse detection. Embedded sensors measure parameters such as applied force, compression depth and airflow. These signals are processed by an onboard microcontroller and transmitted to a mobile application, which serves as the primary interface for real-time feedback, guided instruction, and performance analytics. The use of a mobile platform ensures accessibility, portability, and scalability, allowing users to train independently while maintaining standardized evaluation without requiring continuous instructor supervision. By replacing subjective assessment with quantitative validation, the system improves training consistency and skill acquisition. The solution contributes to broader sustainable development goals by expanding access to quality life-saving training, particularly in locations of limited resources like rural-India. It strengthens community-level emergency preparedness, reduces preventable mortality, and promotes equitable healthcare education, bridging the gap between theoretical knowledge and practical competence.

1. Introduction

1.1 Background and Motivation

Basic life-saving procedures such as CPR and the Heimlich manoeuvre are critical for improving survival during emergencies, especially where timely medical assistance is limited. Incorrect practice may lead to ineffective or potentially dangerous real-life execution. However, many local schools and rural institutions lack access to trained medical professionals, and commercially available professional training manikins are expensive. Practical training often depends entirely on instructor availability and there is no structured self-learning medical simulation system available at low cost. In such cases, students are exposed only to theoretical knowledge and not exposed to the challenges faced during a real-life medical emergency. Traditionally, there is no measurable feedback system to validate the performance of students. Current procedures for evaluation are subjective to the instructor's eye, which this project seeks to address, adding value to even instructor-guided sessions. Our motivation is to provide universal access to quality medical training, regardless of economic and technical background.

1.2 Process Overview

The training process begins with introducing users to specific emergency procedures. Users then perform practical sessions on the manikin, which is embedded with strategically placed electrical sensors to capture parameters (such as compression depth, force and airflow) and simulation of a real-life human being with vibrating pulse. During practice, the system provides detailed real-time feedback through the app dashboard, enabling immediate correction of technique. All performance data are recorded and analysed to generate a final report that includes quantitative feedback, error identification, and actionable suggestions for improvement for each individual learner. The training modules include CPR, force measurement, rescue breathing, and pulse monitoring.

1.3 Flowcharts of Functioning

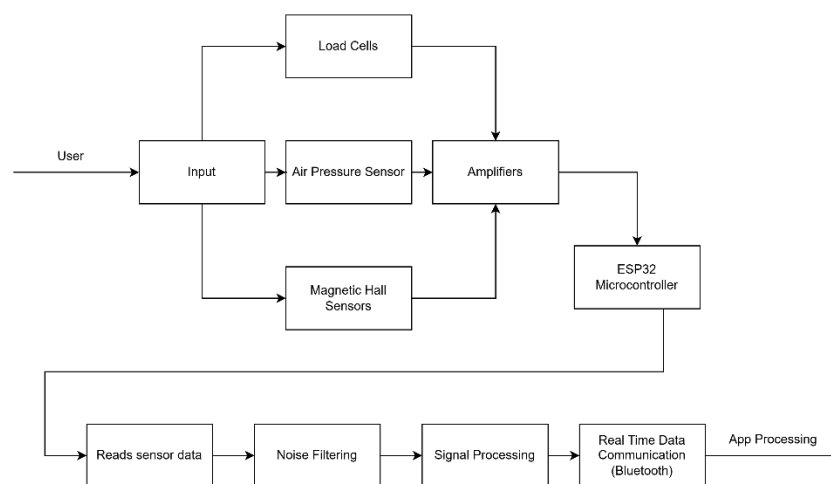


Fig 1.3.1 Sensors and Embedded Systems

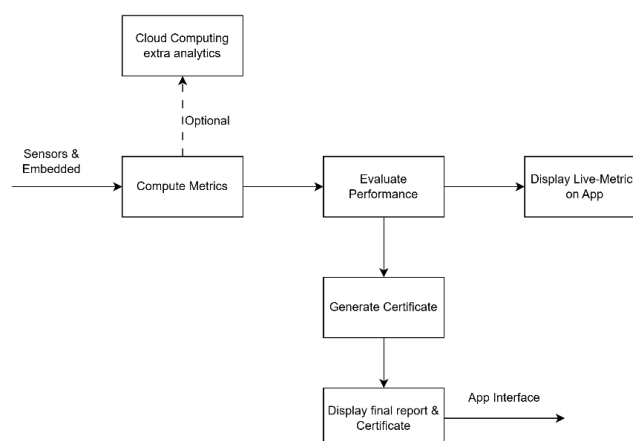


Fig 1.3.2 App Processing

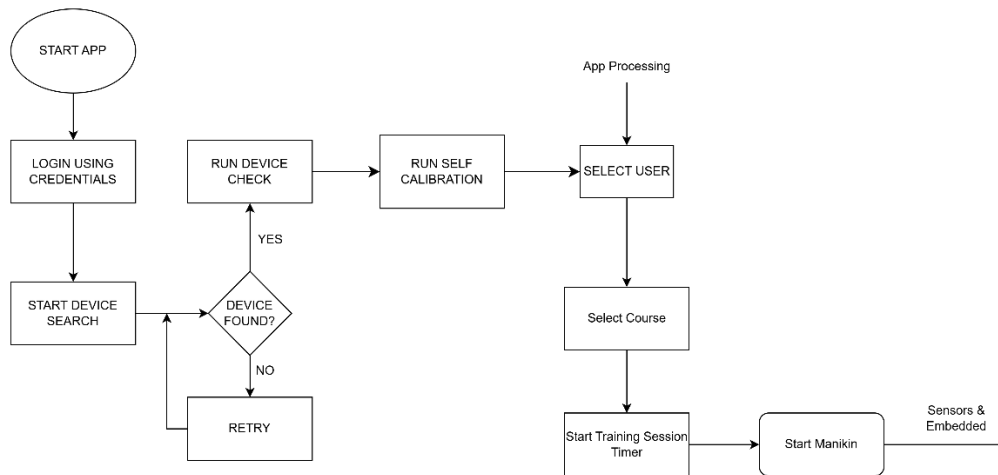


Fig 1.3.3 App Interface System

1.4 Scope of this work

The functionality included in this work is:

A. Electronic System:

1. Measuring the depth of compression.
2. Measuring the force of compression.
3. Data acquisition via sensor reading, noise-filtering, communication with the app through Bluetooth.
4. The charger module, rechargeable battery, and polyfuse to protect the rest of the circuit in the event of overflow of current.
5. Modules to lower voltage to safe and stable limits and provide stable, low-noise power to sensors.

B. App Development:

1. Authentication and User Management
2. Connectivity and Device Integration
3. Core Training Modules
4. Reporting, History & Diagnostics
5. Application Settings and UX

C. Feedback Metrics:

- Training modules mapped out for CPR, Rescue Breathing, and Pulse Checking.
- Mode Selection: Supports both “Training” (for guided instruction) and “Test” modes.
- Interactive Tutorials: A step-by-step tutorial interface structured to accommodate future video content integration.
- Live Session Dashboard: Real-time data visualization screen featuring countdown timers and mock sensor metrics displays.

D. Manikin Construction

Manikin body was built considering the anatomical structure of the body and appendage (head and arms) and an airway. Material was also considered in the construction to give room for compression depth and force measurement. To provide structure and stability a hardstop mechanism was also implemented.

1.5 Literature Review

1.5.1 Medical Metrics

1. Cardiopulmonary Resuscitation (CPR)

Effective CPR is governed by well-established clinical guidelines that define both compression depth and rate. The recommended compression depth for adults is between 5–6 cm, ensuring adequate blood circulation without causing excessive injury. The compression rate should be maintained between 100–120 compressions per minute, with a typical cycle consisting of 30 compressions followed by 2 rescue breaths [\[1\]](#). Proper force application, typically in the range of 40-60 kgf [\[5\]](#), is required to achieve the desired compression depth.

2. Rescue Breathing

Rescue breathing is an essential component of basic life support, aimed at maintaining oxygen supply to vital organs. The recommended tidal volume for each breath is approximately 500-600 mL [\[2\]](#), which corresponds to around 6-7 mL per kilogram of body weight. Each breath should be delivered over approximately one second, ensuring visible chest rise without excessive pressure. Proper airway management is crucial; the nostrils must be closed during ventilation, and if chest rise is not observed, airway obstruction should be suspected. In such cases, repositioning the head to open the airway is necessary before attempting ventilation again. It is also important to limit the number of breaths to avoid hyperventilation, which can adversely affect circulation.

3. Pulse Detection

Pulse assessment is a critical diagnostic step in emergency response, used to determine the need for CPR. For an unresponsive individual, pulse should be checked within 5-10 seconds. The carotid artery is the preferred site due to its stronger and more reliable signal, although peripheral sites such as the wrist may also be used. A normal resting pulse rate ranges between 60–100 beats per minute. In practice, pulse rate is often estimated by counting beats over a 10-second interval and multiplying by six. Failure to detect a pulse within the specified time frame indicates the need to initiate CPR immediately.

Adult* Basic Life Support Algorithm for Health Care Professionals

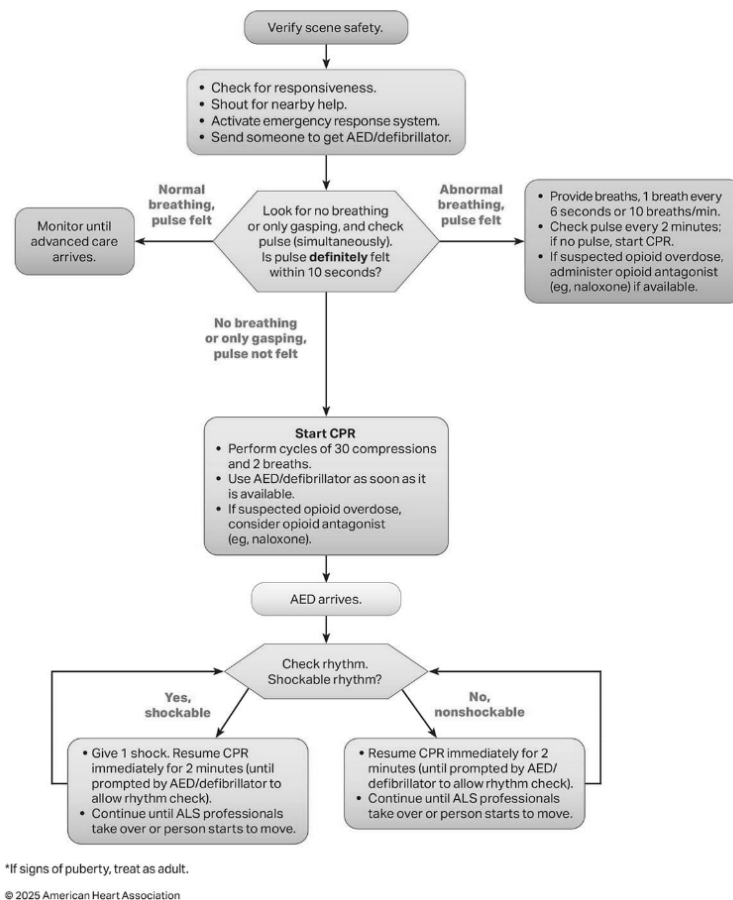


Fig. 1.5.1.1 Basic Adult Life Support Algorithm ^[1]

1.5.2 Important Medical Perspective Considerations

1. Force Distribution and Hand Placement

Accurate hand positioning during CPR is essential to ensure effective compressions and avoid injury. Uneven force distribution can significantly reduce CPR efficiency.

2. Chest Recoil Measurement

Complete chest recoil between compressions is necessary for proper cardiac filling. Monitoring recoil is as important as measuring compression depth.

3. Ventilation Pressure Control

Excessive ventilation pressure can lead to complications. Precise monitoring of airflow pressure is required for realistic training.

4. Airway Patency and Head Positioning

Proper head tilt and airway opening mechanisms are critical for effective rescue breathing but are not currently measured in the system.

5. Realistic Tissue Mechanics

Human chest behaviour involves nonlinear stiffness and damping characteristics, which are difficult to replicate using simple materials. [\[3\],\[4\]](#)

6. Response Time and Decision-Making

In real emergencies, rapid assessment and decision-making are crucial. Training systems should incorporate time-based performance evaluation.

1.6 Market

The target market includes community centres, rural schools, healthcare training institutes, universities, and emergency response training organizations. The system is particularly suited for low-resource settings where access to professional training equipment is limited. Social welfare groups such as NGOs may also benefit with the affordability of the product.

2. Methodology

2.1 Overall

1. Requirement Study: Review clinical guidelines for CPR, breathing, pulse and analyse existing manikin systems.
2. System Design: Define overall architecture (mechanical subsystems, sensors, ESP32, mobile app) and specify measurable parameters and medical metrics for each module. Also think of the overall flow of data.
3. Prototyping: Develop chest compression prototypes, experimentally optimise stiffness and sensor placement. Interface sensors, implement firmware for real time metric extraction, and build a mobile app for live feedback and session logging.
4. Testing & Validation: Conduct practice sessions, calibrate sensors, refine thresholds, and validate feedback accuracy.
5. Integration & Refinement: Integrate all modules, improve robustness, and finalise documentation and demonstration.

2.2 Sensors

The sensing system was designed to capture key physiological parameters associated with emergency medical procedures. Load cells [\[8\]](#) along with HX711 ADC amplifier [\[6\]](#) modules were used for the purpose of measurement of force. A Hall Effect sensor [\[9\]](#) was employed to measure chest compression depth by detecting the displacement of a magnet attached to the moving chest plate. This provides a non-contact and reliable method for tracking compression accuracy.

An air pressure sensor was integrated into the airway channel of the manikin to measure airflow during rescue breathing. This enables estimation of breath adequacy based on pressure variation.

A DC vibration motor was incorporated to simulate a human pulse, providing tactile feedback to the user.

Sensor selection was guided by considerations of cost, availability, ease of integration, and accuracy for training purposes. Testing and validation were performed prior to integration.

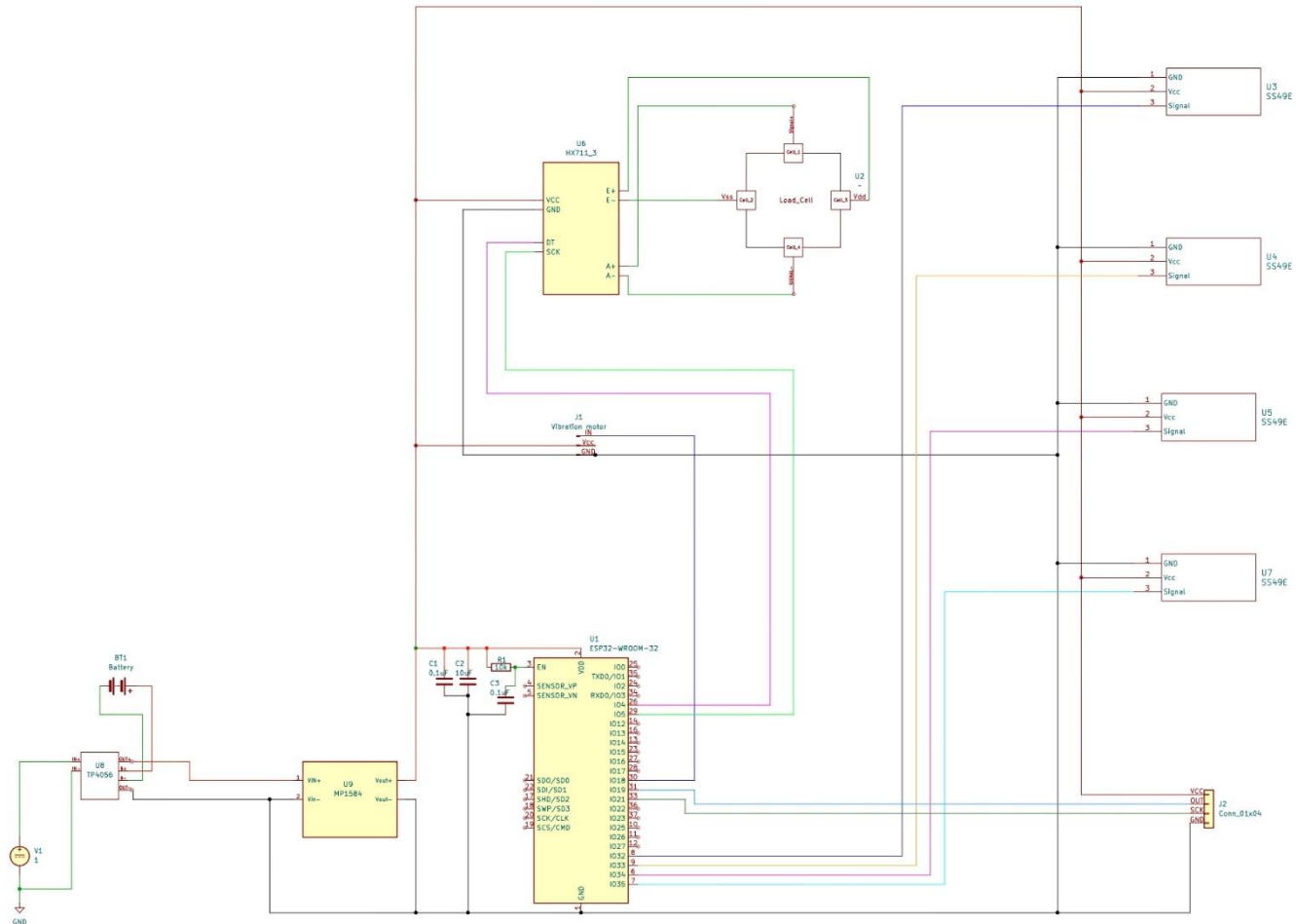


Fig 1.4.1 Schematic of Electronic System

2.3 Embedded System

The embedded system was developed to acquire, process, and transmit sensor data in real time. An ESP32 WROOM microcontroller [\[7\]](#) was selected as the central processing unit due to its integrated Bluetooth capability, low power consumption, and sufficient computational capacity.

Sensor signals were interfaced with the ESP32 and processed using embedded firmware developed in the Arduino IDE environment. To improve signal reliability, a moving average filter was implemented to reduce noise and fluctuations in sensor readings. Calibration algorithms were applied to convert raw sensor outputs into standardized measurement units.

Processed data were transmitted to the mobile application via Bluetooth using the built-in communication libraries. The system architecture ensures low-latency data transfer, enabling real-time feedback during training sessions.

2.3 Power Supply

The system is powered by a 3.7V Lithium-Ion Polymer (LiPo) battery [\[11\]](#), selected for its high energy density and rechargeability.

A charging module with integrated protection circuitry was used to manage battery charging and prevent overcharging, over-discharging, and short circuits. Voltage regulation was achieved using buck converters [\[10\]](#), which step down and stabilize the supply voltage to levels suitable for the microcontroller and sensors.

Additional protection was provided using a polyfuse, which safeguards the system against overcurrent conditions. Capacitors were strategically placed across the circuit to reduce voltage fluctuations and ensure low-noise operation, particularly for sensitive analog sensor readings.

The overall design prioritizes electrical safety, efficiency, and consistent performance during prolonged usage.

Testing and validation were performed prior to integration.

2.4 App Development

App repository, a Flutter-based companion app for the IoT Smart Medical Training Manikin.

Feature Overview

1. Authentication & User Management

- **Signup /Login:** Supports traditional email and password authentication backed by Supabase Auth.
- **Google OAuth:** Integrated `google_sign_in` combined with Supabase ID tokens for seamless social login.
- **Persistent Sessions:** User sessions are remembered and restored across application restarts.
- **Profile Switching:** Netflix-style multi-profile management allowing different users to switch profiles easily (persisted in the Supabase profiles table).

2. Connectivity & Device Integration

- **Bluetooth Classic Setup:** Bluetooth device scanning and connecting capabilities using `flutter_bluetooth_classic_serial`. Features foundational components for SPP/RFCOMM serial streams.

- **Runtime Permissions:** Gracefully handles Android/iOS permission requests (BLUETOOTH, LOCATION, BLUETOOTH_SCAN/CONNECT) depending on OS versions, complete with fallback screens.
- **Pulse Rate Exercise:** Communicates with the ESP32 via Bluetooth to send random BPM numbers. Users are required to guess the BPM and receive feedback through the application.

3. Core Training Modules

- Training modules mapped out for CPR, Rescue Breathing, and Pulse Checking.
- Mode Selection: Supports both “Training” (for guided instruction) and “Test” modes.
- Interactive Tutorials: A step-by-step tutorial interface structured to accommodate future video content integration.
- Live Session Dashboard: Real-time data visualization screen featuring countdown timers and mock sensor metrics displays.

4. Reporting, History & Diagnostics

- Session History: Historical sessions are saved persistently using Supabase, accessible in a comprehensive user-facing list.
- Certification Management: Tracks user certifications and ensures they are safely logged in Supabase.
- PDF Certificate Generation: Ability to generate, download, and share standard PDF certificates signed with a SHA-256 validation code.
- Performance Evaluation: Provides post-session reports, complete with gauge charts, bar charts, correctness scores, and targeted recommendations.
- Manikin Health Diagnostics: Device status overview detailing battery levels, active sensors, and firmware versioning.
- Instructor Admin Dashboard: A dedicated view for instructors detailing multi-tenant trainee metrics pulled from Supabase.

5. Application Settings & UX

- Theming: Full application dark and light mode toggle persistently managed via shared preferences.
- Sensory Settings: Customizable toggles for sound cues, haptics, and notifications.
- Home Dashboard: Quick actions and app entry point interface.

2.5 Manikin Body Construction

To create a realistic and functional CPR training manikin, materials were selected based on their tactile properties and their ability to simulate human physiological responses.

1. Anatomical Structure and Appendages

- Limbs (Arms): High-density polyethylene foam tubes (pool noodles) were utilized for the arm structures. These provide a lightweight yet semi-rigid form. The core will be hollowed so that the core of the foam serves as a natural conduit for the internal motor wiring, ensuring a clean and protected integration of electronic components.

- **Head and Airway:** The head was sculpted using foam. To simulate the oral cavity for rescue breathing, a specialized intake valve (modified straw/pen refill) was integrated into the lips to facilitate air passage into the internal sensor system.

2. Compression Simulation (Chest Cavity)

The chest assembly is designed to provide realistic resistance and depth feedback during CPR practice:

- **Variable Resistance:** A dual-layer system of soft and rigid foam was implemented to simulate human chest compliance. The soft foam mimics the initial skin and tissue compression, while the high-density hard foam regulates the depth of the compression to meet clinical training standards.
- **Compression Depth Feedback:** This layered approach allows the sensors to accurately measure the travel distance, ensuring the trainee applies the correct force required for effective chest compressions.

3. Internal Component Housing and Protection

Electronic Integrity: To safeguard the microcontrollers and force sensors, a structural chassis (hardstop box) was integrated into the core of the manikin.

The Hardstop Mechanism: This box serves two purposes:

- It acts as a mechanical hardstop to prevent over-compression from damaging sensitive internal electronics.
- It provides a stable, level base for the sensors to rest against, ensuring consistent data acquisition during training sessions.

2.6 Materials Used

Table 2.6.1 Materials Used

S. No	Item	Purpose	Net Price (Rs)
1	Linear Hall Effect Sensor Module	Measure compression depth	120
2	ESP32-S3 Dev Module	Processing + BLE communication	378
3	3.7V 1500mAh 1S LiPo Battery	Primary power source	445
4	TP4056 Charging Module (with protection)	Battery charging & protection	14

5	High Efficiency Buck Converter (MP1584)	Primary voltage regulation	88
6	1A Polyfuse	Overcurrent protection	12
7	Slide Switch	Main power control	3
8	DC Vibration Motor (3–5V)	Simulated pulse feedback	73
9	AO3400 / IRLZ44N MOSFET	Motor driver control	52
10	1N5819 Schottky Diode	Flyback protection for motor	2.67
11	0.1 μ F Ceramic Capacitors	Local IC decoupling	15
12	10 μ F Capacitors	Analog rail stabilization	13.75
13	100 μ F Electrolytic Capacitors	Motor rail stabilization	18
14	100k Ω Resistors (Battery Divider)	Battery voltage monitoring	1
15	1k Ω Resistor	Hall sensor RC filter	10
16	Precision Gain Resistors (0.1%)	Set INA333 gain accurately	40
17	PCB Prototype Board	Analog-digital separation & integration	61
18	Foam (Pack of 10)	Chest travel	448
19	Foam Cushion	Chest structure	445
20	Pool Noodle	Limbs	527
21	Pack of magnets	For hall sensor	269
22	Hx710B Air Pressure (0–40KPa) Sensor Module with Straight Pin	To measure differential pressure	135
23	Miscellaneous (wires, cutter, tape, glue, etc)		500
TOTAL	Total cost		3670.42

3. Results

3.1 Commercially Available Costs

Commercially available medical training manikins typically range from Rs. 30,000 to Rs. 50,000, with prices going up to even Rs 1.6 lakhs, depending on features and build quality.

3.2 Overall Project Cost

The total cost of the developed system is approximately Rs. 3670, which is significantly lower than existing commercial alternatives. This represents a cost reduction of more than 10 times, demonstrating the affordability and potential for wider adoption, especially in resource-limited settings.

Despite the lower cost, the system successfully incorporates essential features such as sensor-based measurement, real-time feedback, and mobile application integration. While high-end systems may offer greater anatomical realism and advanced simulation capabilities, the core objective of training users in correct CPR technique is effectively addressed by the proposed design.

The use of readily available components and a modular design approach contribute to both affordability and ease of replication. This makes the system particularly suitable for large-scale deployment in educational institutions and low-resource environments.

The results indicate that meaningful training outcomes can be achieved without the need for expensive equipment, highlighting the potential for broader accessibility.

3.3 Medical Metrics

Parameter	Recommended Standard	Method of Measurement
Compression Depth	5 – 6 cm	Hall Effect Sensor + Magnet Displacement
Compression Rate	100 – 120 compressions/min	Time-based app tracking
Compression Force	40 – 60 kgf (~390 – 590 N)	Load Cell + HX711
Rescue Breathing Volume	500 – 600 mL	Air Pressure Sensor
Pulse Detection	60 – 100 bpm	User interaction + app input
Data Transmission	Real-time	Bluetooth (ESP32)
Power Stability	Continuous operation	Battery + regulation circuit

4. Discussion

4.1 Future Scope

1. **Better Chest Simulation**
The chest mechanism can be improved using springs or similar systems to more closely match how a real human chest behaves during CPR.
2. **Improved Pulse Monitoring System**
The pulse feature can be made more realistic, and additional sensors can be added to check whether the trainee is detecting the pulse correctly according to what the trainee reports.
3. **AED Training Module**
The system can be expanded to include training for Automated External Defibrillators (AEDs) to simulate real emergency situations.
4. **Additional First Aid Modules**
Training for scenarios such as fracture handling and splint application can be added to make the system more comprehensive.
5. **Hand Placement Detection**
Sensors can be included to check whether the user is placing their hands correctly during CPR.
6. **Stronger and More Durable Design**
The manikin structure and material can be improved using better materials to increase durability while keeping costs low.
7. **User progress can be used to train an AI Chatbot to guide user for future sessions, i.e. as a context-aware guide.**

4.2 Limitations of Current Work

1. **Limited Realism**
The manikin does not fully replicate the structure and behaviour of the human body, which may affect real-world accuracy.
2. **Basic Feedback Capabilities**
The current design does not verify hand placement and provides limited validation for pulse detection.
3. **Durability and Scalability**
The prototype model is not yet fully optimized for long-term use.

5. Conclusion

Cost Analysis:

Commercially available medical training costs range from Rs 30,000 to Rs 50,000, even reaching upto Rs. 1.6 lakhs, at the time of the preparation of this report. Our current product costs Rs 3670 which is cost reduction of more than 10 times.

Cost Reduction:

The overall cost of the system is primarily driven by the construction of the manikin body. With better sourcing of materials and optimized design, this cost can be further reduced. The electronic components' net price can be dropped further when the project is scaled and components are purchased in bulk.

Highest Effort Areas:

The most effort was required in developing the manikin body and integrating the sensors with the mobile application. Despite this complexity, a clear workflow has been established in the product preparation, testing, validation enhance the scalability of the product, and the application has been designed to support future additions and improvements, along with ease of user experience with trainee and instructor model.

Proof of Concept:

The current prototype successfully demonstrates the feasibility of a low-cost, sensor-enabled medical training system with real-time feedback.

6. References

[1] "Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care" American Heart Association (2025)

[2] Baskett, P., Nolan, J., Parr, M., (1996) "Tidal volumes which are perceived to be adequate for resuscitation" Elsevier 31-3, June 1996

DOI: 10.1016/0300-9572(96)00994-X

[3] A. A. Stanley, S. K. Healey, M. R. Maltese, K. J. Kuchenbecker, "Recreating the feel of the human chest in a CPR manikin via programmable pneumatic damping" IEEE Haptics Symposium (HAPTICS), Vancouver (2012)

DOI: 10.1109/HAPTIC.2012.6183767

[4] Lim, H. et al., "Variable Stiffness and Damping Mechanism for CPR Manikin to Simulate Mechanical Properties of Human Chest" IEEE Health Med. (2024) Jul 16;12:542-549.

DOI: 10.1109/JTEHM.2024.3429422

[5] Dorph E. et al., "Compression force–depth relationship during out-of-hospital cardiopulmonary resuscitation" Resuscitation, 2007.

- [6] Avia Semiconductor, “24-Bit Analog-to-Digital Converter (ADC) for Weigh Scales: HX711,” Datasheet
- [7] Espressif Systems, "ESP32-WROOM-32 Datasheet, Version 3.6", Espressif Systems (Shanghai) Co., Ltd., 2025.
- [8] CZL635 Micro Load Cell (0–20 kg), 3134 – Micro Load Cell Datasheet, May 13, 2011.
- [9] Honeywell, “SS39ET/SS49E/SS59ET Series Linear Hall-Effect Sensor ICs Datasheet”, Dec. 2015.
- [10] Monolithic Power Systems, “MP1584: 3A, 1.5 MHz, 28 V Step-Down Converter Datasheet”, Rev. 1.0, Aug. 8, 2011.
- [11] UTD Semiconductor Co., Ltd., TP4056: 1 A Standalone Linear Li-Ion Battery Charger Datasheet

7. Supplementary Material

Sample Code Used For Integrating and Communicating

```
#include "BluetoothSerial.h"
#include <ArduinoJson.h>
BluetoothSerial SerialBT;
void setup() {
    Serial.begin(115200);
    SerialBT.begin("Manikin_BT");
    Serial.println("Bluetooth Started. Ready to pair.");
}
void loop() {
    // 1. Simulate/Get averaged data
    int sensor1 = analogRead(32);
    int sensor2 = analogRead(33);
    bool motorStatus = true;
    // 2. Create the JSON document
    StaticJsonDocument<200> doc;
    // 3. Add data
    doc["s1"] = sensor1;
    doc["s2"] = sensor2;
    doc["motor"] = motorStatus;
    doc["id"] = "manikin_01";
    // 4. Send over Bluetooth
    serializeJson(doc, SerialBT);
    SerialBT.println(); // Signal end of JSON object
    // 5. Fix: Correct way to print to Serial Monitor
    Serial.print("Sent data, S1 value: ");
    Serial.println(sensor1);

    delay(500);
}
```

```

Future<void> startScan() async {
  final bluetooth = FlutterBluetoothClassic();
  state = state.copyWith(isScanning: true, devices: []);

  await bluetooth.startDiscovery();
  await Future.delayed(Duration(seconds: 12));
  await bluetooth.stopDiscovery();
  final devices = await bluetooth.getDiscoveredDevices();

  state = state.copyWith(
    isScanning: false,
    devices: devices
  );
}

void stopScan() {
  state = state.copyWith(isScanning: false);
}

```

```

/// Device connection state using the native BluetoothDevice.
class DeviceState {
  final bool isScanning;
  final List<BluetoothDevice> discoveredDevices;
  final BluetoothDevice? connectedDevice;
  final int batteryLevel; // 0-100
  final String connectionQuality; // Excellent, Good, Fair, Poor
  final bool isConnecting;

  const DeviceState({
    this.isScanning = false,
    this.discoveredDevices = const [],
    this.connectedDevice,
    this.batteryLevel = 0,
    this.connectionQuality = 'N/A',
    this.isConnecting = false,
  });
}

```

```

void _listenToDeviceData(FlutterBluetoothClassic bluetooth) {
  // Cancel any old subscriptions just in case
  _dataSubscription?.cancel();

  // Listen to the raw byte stream coming from the ESP32
  _dataSubscription = bluetooth.onDataReceived.listen((data) {

    // Convert the raw bytes (Uint8List) into a String
    final String incomingMessage =
    utf8.decode(data.data).trim();
  });
}

```



```

        print('Received from ESP32: $incomingMessage');

        // 3. PARSE THE STRING AND UPDATE STATE
        if (incomingMessage.startsWith("BAT:")) {
            // Extract the number after "BAT:"
            final String batString = incomingMessage.substring(4);
            final int? parsedBattery = int.tryParse(batString);

            if (parsedBattery != null) {
                // Update the UI instantly
                state = state.copyWith(batteryLevel: parsedBattery);
            }
        }

        // You can add more parsers here if you send "QUAL:80" or
        "TEMP:98"
    });

```

```

void loop() {
    unsigned long now = millis();

    // Throttle to LOOP_INTERVAL_MS
    if (now - lastLoopMs < LOOP_INTERVAL_MS) return;
    lastLoopMs = now;

    // ---- Read incoming BT (line-delimited JSON) ----
    while (SerialBT.available()) {
        char c = (char)SerialBT.read();
        if (c == '\n' || c == '\r') {
            btBuffer.trim();
            if (btBuffer.length() > 0) {
                Serial.print("[BT RX] "); Serial.println(btBuffer);
                handleBTMessage(btBuffer);
                btBuffer = "";
            }
        } else {
            btBuffer += c;
        }
    }

    // ---- Session timeout ----
    if (sessionActive && sessionTimedOut()) {
        Serial.println("[SESSION] Timed out – stopping.");
        endSession();
        return;
    }

    // ---- Test mode: ramp pulse after CPR phase ----

```

```

    if (currentMode == MODE_TEST && sessionActive) {
        if ((now - sessionStartMs >= TEST_PULSE_RAMP_DELAY_MS) &&
pulseFreqBPM == 0) {
            Serial.printf("[TEST] Pulse ON → %d BPM\n",
testPulseTargetBPM);
            setPulse(testPulseTargetBPM);
        }
    }

    // ---- Motor pulse generation ----
    updatePulseMotor();

    // ---- Send sensor data at SEND_INTERVAL_MS ----
    if (sessionActive && (now - lastSendMs >= SEND_INTERVAL_MS)) {
        lastSendMs = now;
        switch (currentMode) {
            case MODE_COURSE_CPR:    sendCPRData();    break;
            case MODE_COURSE_PULSE:  sendPulseData();  break;
            case MODE_TEST:          sendTestData();   break;
            default: break;
        }
    }
}

```